

Learner Guidance

Estimation · How can an estimate help us decide whether an answer makes sense?

You can pause and return.

You do not need to complete everything in one sitting. Using support, including a Reconnection Route, is part of learning.

What are we exploring?

You will use rounded, friendly numbers to predict the likely size of a calculation. An estimate is not pretending to be exact. It helps you notice whether an answer is sensible.

Your Stepping Stones

- round values to friendly numbers, usually 1 significant figure;
- calculate mentally with the rounded values;
- compare an estimate with an exact result;
- use the estimate to identify a suspicious answer;
- explain whether an estimate is useful for its purpose.

Before you begin

- You may use the in-lesson Scratchpad, paper, a physical whiteboard or another recording method.
- A calculator is useful for checking the exact result after you have estimated.
- Choose a quieter space, reduce the screen brightness or take a movement break if that helps.

Reconnection Routes

The Reconnection Routes button is near the top of the lesson. Choose a short recap only if place value, rounding, magnitude or an operation is blocking the current task. Each route includes a Return to Estimation button.

A possible way through the lesson

1. Connection: choose which possible answer has a sensible size.
2. Movement: use the Estimation Laboratory. Round both values to 1 significant figure and calculate with the friendly values.
3. Reflection: compare the estimate with the exact result and decide whether the estimate did its job.
4. Creativity: design a calculation with one plausible and one suspicious answer.
5. Rest: pause and retrieve the route round → calculate → check.
6. Nutrition: notice how estimation nourishes judgement, confidence and independence.
7. Final Task: investigate two calculations and explain what the estimates helped you decide.

Timings are invitations, not deadlines.

You may stop after a meaningful piece of thinking and continue later.

Using the Estimation Laboratory

8. Choose or enter two values and an operation.
9. Read the rounded calculation shown by the page.
10. Enter your estimated result.
11. Check your estimate.
12. Compare the estimate with the exact result.
13. Record what the comparison tells you.

Practice Companion and Scratchpad

Practice Companion

The six questions include addition, subtraction, multiplication, division, decimals and a suspicious calculator display.

Universal worked-solution release rule

For every question independently: Socratic prompt used + Hint used + two non-empty answers entered and checked = Worked solution released. The conditions begin again for each question.

The button tells you what remains to do. A Supporting Adult should not give you the complete worked solution while it is locked.

Estimation Organiser Scratchpad

- Original values: copy the calculation.
- Rounded values: write friendly values.
- Estimate: calculate mentally.
- What I notice: compare with the exact result or identify a suspicious answer.

The Scratchpad is optional. Erasing or clearing removes your marks while preserving the selected mathematical background.

When you feel stuck

- Estimate only the likely size first.
- Use one Reconnection Route, then return.
- Say the calculation aloud using friendly values.
- Try a different recording method.
- Pause and come back.

Showing what you understand

You may show understanding through typed calculations, spoken explanation, annotations, a Scratchpad image, paper working or a combination.

A useful explanation includes

- the original values;
- the rounded values;
- the estimated result;
- the exact result when appropriate;
- a judgement about reasonableness;
- a sentence explaining whether the estimate was useful.

Short self-review

- | | | | |
|----------------------------------|---------------------------------------|---|--|
| ☐ Not yet | ☐ With support | ☐ Secure for now | I can choose friendly rounded values. |
| ☐ Not yet | ☐ With support | ☐ Secure for now | I can estimate using the four operations. |
| ☐ Not yet | ☐ With support | ☐ Secure for now | I can judge whether an exact answer is reasonable. |
| ☐ Not yet | ☐ With support | ☐ Secure for now | I know when I need a Reconnection Route. |

A valid finishing point

Stop when you have made one useful estimation decision and can name your next connection.
Incomplete work is information, not failure.